

Toward Principled Threshold Selection for Blink Region Detection in Biological Signals: A Cross-Dataset Benchmark with LLM-Guided Candidate Derivation

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- **ABSTRACT** This study evaluated an epoch-aware pipeline for EEG blink-region detection that avoided estimating detection thresholds from concatenated blink-containing and blink-free epochs. The three-stage pipeline first screened non-overlapping epochs by peak-to-peak amplitude using an autoreject-inspired cross-validation criterion, then estimated a sample-level threshold from the suspicious-epoch pool using either a mean-based or median/MAD-based estimator with a fallback for small pools, and finally extracted blink regions and mapped them to epoch-local coordinates. BLINKER-concat, MNE-annot, Proposed-Mean, and Proposed-Med were compared on the Raja and Cao2018 naturalistic driving-EEG corpora using event-level overlap matching and macro-averaged F_1 . At 30 s, Proposed-Med achieved a pooled macro- F_1 of 0.8403, significantly exceeding BLINKER-concat at 0.7256 and MNE-annot at 0.6086. Proposed-Mean and Proposed-Med were statistically indistinguishable ($p = 1$), which indicated that the gain came from epoch screening rather than the choice of Stage-B estimator. The channel-selection ablation further showed that a single frontopolar electrode was sufficient for the strongest single-channel detection on both corpora. Finally, pooled macro- F_1 varied by only 0.0056 across epoch durations from 10 s to 120 s, and no tested duration differed from the 30 s reference after Bonferroni correction.
- **INDEX TERMS** blink detection, electroencephalography, epoch-aware detection, naturalistic driving, threshold selection

I. INTRODUCTION

Blinking is a physiological process that involves the closure and reopening of the eyelids [1]. Its primary functions are to protect the ocular surface, maintain corneal moisture, and support visual comfort [1]. Beyond these physiological functions, blink behaviour has been increasingly recognised as an important indicator in fatigue-driving research [2, 3]. Variations in blink-related features, such as blink frequency, blink duration, eye-closure duration, and blink patterns, can reflect changes in alertness and cognitive state [2, 4]. As fatigue increases, drivers may exhibit slower or prolonged blinks, which can indicate reduced vigilance and a higher risk of impaired driving performance [2]. Therefore, blink behaviour provides a useful and non-invasive marker for monitoring driver fatigue [3, 5].

In the neuroscience study of fatigue driving, blink activity can be captured through external methods, such as camera-based eye monitoring, or through electrophysiological recordings, particularly electroencephalography and electrooculography [3, 5]. Although EEG is mainly used to mea-

sure brain activity, blink-related components are often present in EEG signals because eyelid movements and eye movements generate strong electrical potentials [6, 7]. These ocular activities are especially prominent in frontal EEG electrodes due to their close proximity to the eyes [6, 8]. As a result, blink information can be extracted from EEG recordings, either by analysing the ocular components embedded in the EEG signal or by combining EEG with EOG-based measurements [5, 9]. This makes EEG/EOG recordings valuable not only for studying neural activity during fatigue but also for identifying blink-related markers associated with drowsiness [2, 10].

It is important to distinguish between two major perspectives on blink activity in the existing literature [3, 6]. In many EEG studies, blink-related activity is treated as an artifact because it contaminates neural signals and may interfere with the interpretation of brain activity [6, 7]. In such cases, blink components are commonly removed during EEG preprocessing [6, 11]. However, in fatigue-driving studies, blink activity can also be treated as meaningful information rather than

noise [3, 5]. From this perspective, blink events are identified, extracted, and analysed as potential indicators of driver fatigue [5, 10]. Blink detection methods can generally be classified into machine-learning-based, morphology-based, and thresholding-based approaches [9, 12]. Machine learning methods rely on trained models to classify blink events, while morphology-based methods identify blinks based on their characteristic waveform shape [13, 14]. Thresholding-based methods, in contrast, detect candidate blink events when the signal amplitude exceeds a certain threshold value, as blink activity typically produces large-amplitude deflections, especially in frontal channels [8, 15].

Thresholding is one of the most straightforward and widely used approaches for detecting blink activity from EEG or EOG signals [15, 16]. The simplicity of thresholding makes it computationally efficient and easy to implement, particularly in real-time fatigue-detection systems [16, 17]. Existing approaches range from single- or few-channel detectors that apply amplitude thresholds with rule-based decision logic to hybrid pipelines that combine thresholding as an initial detection step with additional candidate-refinement mechanisms [8, 9, 12, 14–16, 18–24]. Despite these developments, selecting an appropriate threshold remains challenging because blink amplitude varies across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. Blink amplitude can vary across participants, electrode locations, recording devices, signal quality, and experimental conditions [20, 24]. A threshold that is too low may produce false detections due to noise or other artifacts, whereas a threshold that is too high may miss genuine blink events [16, 17]. This backdrop explains why simple one-value global thresholds can be attractive, yet brittle. Therefore, adaptive and robust threshold selection remains a critical consideration for reliable blink detection, particularly in fatigue-driving applications [5, 10].

In many fatigue-driving studies, EEG recordings are segmented into epochs that correspond to specific task events or analysis windows [25]. This epoch-structured setting introduces additional challenges for threshold-based blink detection [16, 17]. When thresholds are estimated from the entire recording, they may be biased by the presence of clean epochs that contain few or no blinks, which may distort threshold estimates and suppress blink detection in epochs where blinks are more prevalent [11, 24]. As a result, naive application of thresholding in an epoch-structured context may risk under-detection of blinks, which undermines the reliability of fatigue assessments based on blink metrics [5, 10]. Addressing this challenge requires developing blink detection methods that explicitly account for the epoch structure and can adapt threshold estimation to the specific characteristics of each epoch or subset of epochs, rather than relying on a global threshold derived from the entire recording [9, 24].

This paper makes several contributions to blink detection in fatigue driving research. First, we propose a double thresholding blink detection pipeline comprising epoch screening, sample level threshold estimation, and blink region

extraction. To our knowledge, this is among the first frameworks to integrate these three stages explicitly to address the challenges associated with epoch structured EEG recordings. Second, we systematically compare the proposed pipeline against baseline methods, including BLINKER concat [8] and MNE annot. To support reproducibility and facilitate further research, we also developed an automated pipeline for extracting ocular indices from EEG recordings and made it freely available as a Python toolbox called *pyblinker* (TODO). The toolbox is intended to facilitate reproducible blink analysis and encourage more consistent reporting of ocular indices in EEG based fatigue research.

The remainder of the paper is organized as follows. Section Methods and materials describes the steps and methods used to extract blinks from EEG data, as well as the datasets we used to validate the results. Section Results presents the results of running the BLINKER toolbox on a large and varied collection of EEG data. Section The BLINKER software briefly introduces the toolbox, referring readers to the more detailed online documentation and downloads. Section Visual verification of BLINKER output describes our manual video verification process and some interesting results we obtained when comparing subject behavior captured in video with blink signals captured by EEG. Section Discussion offers some concluding remarks and discusses limitations of the method.

II. METHOD AND MATERIALS

a: Processing pipeline overview.

The proposed pipeline is an epoch-aware, threshold-based pipeline that transforms a continuous multichannel EEG recording into a non-overlapping epoch through a fixed sequence of five operations. The Figure 1 summarises this sequence as a flow chart, from the preprocessing, epoching, and the double thresholding which is the proposed method, through three detection stages (A, B, and C) which we will discuss in detail in section.

b: Preprocessing

First, the continuous recording is downsampled to a common sampling grid of 100 Hz. Standardising the sampling rate makes the amplitude thresholds and the minimum-duration gate comparable across recordings acquired on different hardware, and reduces the per-sample computational cost of the downstream stages without discarding any information needed to resolve a blink, whose spectral content lies well below the new Nyquist frequency.

Second, a zero-phase band-pass filter from 1.0 Hz to 20.0 Hz is applied which removes slow drifts and DC offsets at the low end and high-frequency muscle and line noise at the high end, isolating the low-frequency, high-amplitude deflection that characterises an eye blink.

Third, each filtered recording is split into non-overlapping, fixed-length epochs. Epoching is what makes the thresholds *adaptive*, rather than deriving a single level for the entire ses-

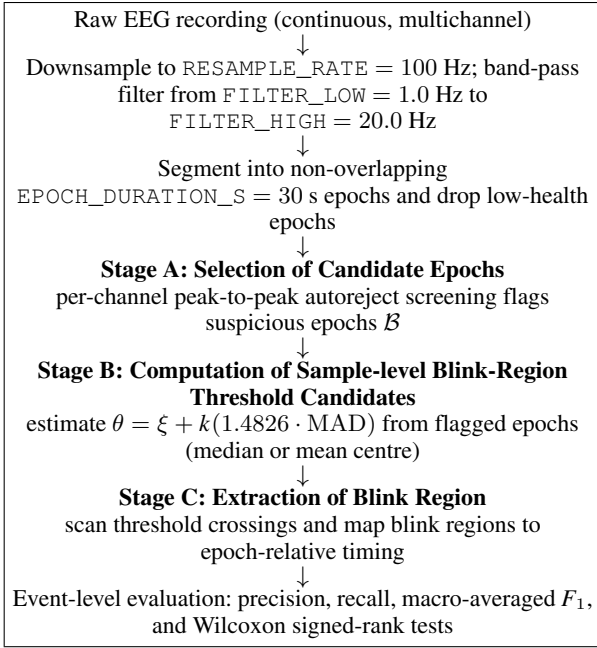


FIGURE 1. Per-epoch blink-detection pipeline. The continuous recording is downsampled and band-pass filtered, segmented into fixed-length epochs with low-health epochs removed, and then passed through the three detection stages: Stage A screens suspicious (blink-heavy) epochs by peak-to-peak amplitude, Stage B estimates a robust sample-level amplitude threshold from those epochs, and Stage C localises blink regions by scanning threshold crossings. Detections are finally scored against the ground truth at the event level.

sion, the pipeline treats each short epoch as a local statistical context.

c: Selection of Candidate Epochs

The first stage, Stage A, of the proposed pipeline performs an initial epoch-level screening using an *autoreject*-inspired threshold-selection procedure to identify epoch that are most likely to contain prominent blink-related activity. Rather than relying on a single channel, the screening decision integrates channel-level information across multiple EOG/ EEG channels, allowing an epoch to be selected when abnormal amplitude excursions are detected in one or more channels. At this stage, the objective is not to localize individual blink samples, but to identify candidate epochs containing prominent amplitude activity. This multi-channel screening strategy therefore narrows the search space to potentially blink-related epochs for more detailed sample-level analysis in the subsequent detection stages.

Let $X \in \mathbb{R}^{N \times C \times T}$ denote the epoched multichannel signal after preprocessing, where N is the number of valid epochs, C is the number of channels, and T is the number of samples per epoch. The sample at epoch $i \in \mathcal{E} = \{1, \dots, N\}$, channel $c \in \mathcal{C} = \{1, \dots, C\}$, and within-epoch time index $t \in \{1, \dots, T\}$ is denoted by $x_{i,c,t}$.

For each epoch and channel, an amplitude-range statistic is computed using the peak-to-peak value

$$a_{i,c} = \max_{1 \leq t \leq T} x_{i,c,t} - \min_{1 \leq t \leq T} x_{i,c,t}.$$

Large values of $a_{i,c}$ indicate that the corresponding epoch contains a high-amplitude excursion in channel c , a pattern consistent with ocular or other transient artifacts. Because signal amplitudes may vary across channels, participants, and recording conditions, a fixed peak-to-peak threshold is not imposed. Instead, a channel-specific threshold τ_c^* is calculated from each of the channel's. Candidate thresholds are taken from the empirical range of peak-to-peak amplitudes for that channel, denoted $\mathcal{Q}_c = \{a_{i,c} : i \in \mathcal{E}\}$, and are evaluated by a cross-validation criterion that measures the discrepancy between retained training epochs and validation epochs.

After the channel-specific thresholds, τ_c^* , have been estimated, the peak-to-peak amplitude of each epoch is evaluated separately for each EEG channel against its corresponding threshold. An epoch is retained as a candidate when the amplitude in at least one channel exceeds the threshold τ_c^* defined for that channel. Thus, the epoch-level screening decision incorporates information across multiple channels using a union rule, rather than relying on a single predefined channel. The resulting set of candidate epochs is defined as The set of suspicious epochs is therefore

$$\mathcal{B} = \{i \in \mathcal{E} : b_i = 1\}.$$

where b_i denotes the epoch-level screening outcome indicating that epoch (i) satisfies the selection criterion. These candidate epochs represent high-amplitude segments that may contain prominent blink-related activity and are retained for subsequent sample-level blink-region detection.

d: Computation of Sample-Level Blink-Region Threshold Candidates

Stage B estimates channel-specific sample-level amplitude thresholds using signal samples drawn from the candidate epochs $\mathcal{B}$ identified in Stage A. The estimation combines a measure of central tendency with a robust measure of dispersion and a sensitivity parameter, allowing the threshold to adapt to the amplitude characteristics of each channel while limiting the influence of isolated extreme samples.

Let $\mathcal{S}$ denote the set of epochs used for estimating the sample-level threshold. In the primary analysis, $\mathcal{S} = \mathcal{B}$, so that the estimate is driven by epochs already identified as likely to contain blink-related activity.

For each channel c , all samples from the selected epochs are pooled into a one-dimensional empirical sample set

$$\mathcal{Y}_c = \{x_{i,c,t} : i \in \mathcal{S}, 1 \leq t \leq T\}.$$

The sample-level threshold is estimated using a robust location-plus-dispersion form. Let ξ_c be the channel-specific center of the sample distribution. The default choice is the median,

$$\xi_c = \text{median}_{y \in \mathcal{Y}_c} y,$$

It should be note that, the arithmetic mean may be substituted when a mean-centered threshold is required for comparison.

The dispersion is estimated by the median absolute deviation (MAD),

$$\text{MAD}_c = \text{median}_{y \in \mathcal{Y}_c} |y - \text{median}_{u \in \mathcal{Y}_c} u|,$$

and is rescaled to a standard-deviation-compatible quantity by

$$\sigma_c^{\text{rob}} = 1.4826 \text{ MAD}_c.$$

The constant 1.4826 is the usual normal-consistency factor for the MAD.

For a sensitivity parameter $k > 0$, the resulting sample-level blink-region threshold candidate for channel c is

$$\theta_c = \xi_c + k\sigma_c^{\text{rob}}.$$

The set

$$\Theta = \{\theta_c : c \in \mathcal{C}\}$$

constitutes the channel-wise threshold candidates passed to the blink-region extraction stage.

A. EXTRACTION OF BLINK REGIONS

The final stage applies the sample-level thresholds $\Theta = \{\theta_c\}_{c=1}^C$ obtained in Stage B to obtain channel-wise blink regions. Specifically, blink regions are searched for within each valid epoch $i \in \mathcal{E} = \{1, \dots, N\}$ and each channel $c \in \mathcal{C}$, by comparing the samples $x_{i,c,t}$ of that epoch and channel against the threshold candidate θ_c of the same channel.

The final output of this stage is therefore a channel-indexed collection of blink regions, each represented by an epoch identifier, a channel label, an onset time, and duration.

B. EPOCH SEGMENTATION AND DETECTION PIPELINE

C. DATASETS

Two naturalistic driving-EEG corpora (i) Stage-A epoch selection F_1 . The Raja corpus comprised 46 complete sessions recorded with a 128-channel EGI system (ANT Neuro), referenced to CPz under a modified international 10-20 layout, during a simulated lane-keeping driving task; ground-truth blink annotations were derived from human EAB/EOC labelling. The channel scope was dataset specific: for Raja, the analysis was restricted to a 32 channels as. Specifically, we only retain the channel

The Cao2018 corpus (ii) Number of channels for Stage A. The Cao2018 corpus (iii) False-positive epoch rate on driving flagged. corpus reported by [25], in which 32-channel EEG was recorded during an approximately 90-minute lane-keeping task; 58 complete sessions were used, and 4 non-complete sessions were excluded. For Cao2018, all 32 EEG channels were provided to the pipeline.

D. DETECTOR CONFIGURATIONS

Four detector configurations were compared (iv) Best-identical stability. preprocessing and epoching. The proposed methods were Proposed-Med, which implemented the three-stage pipeline with a median centre, and Proposed-Mean, which used a mean centre. The conventional baselines (v) BLINKER robustness. concat and MNE-annot.

E. EXPERIMENT OVERVIEW

The experiments follow a deliberate logical progression. Experiment 1 first selects the best Stage-A channel-selection configuration; Experiment 2 then benchmarks that chosen configuration against competing detectors; and Experiment 3 stress-tests the resulting pipeline against the one analysis choice most likely to be imposed by the experimental paradigm rather than by the detector, namely epoch length. Fixing the channel configuration first, before any baseline comparison, ensures that the headline comparison in Experiment 2 uses a principled spatial selection rather than an arbitrary default, and that the robustness check operates on the same ideal configuration. Each experiment is summarised below in terms of its purpose, motivation, and objective.

a: Experiment 1 – Channel-selection strategies.

Purpose: determine which Stage-A channel-selection group yields the best detector, evaluated on both datasets (Raja and Cao2018) and under both estimators (median and mean), before any other experiment is run. *Motivation:* the current default rule flags an epoch whenever *any* channel over *all* channels exceeds its peak-to-peak threshold, which can be triggered by a single noisy non-ocular channel and inflate the number of flagged epochs; a more selective spatial group may screen blinks more cleanly. *Objective:* run the complete Stage A→B→C pipeline on each candidate channel group and select the best group by weighing multiple criteria rather than a single metric. The decision criteria are:

- (i) End-to-end event-level F_1 . The end-to-end event-level F_1 of the full pipeline when restricted to the group; this is the ultimate quantity the detector is optimised for and is therefore the primary criterion.
- (ii) The precision and recall of the suspicious-epoch selection itself, scored against epoch-level ground truth (an epoch is positive if it contains at least one annotated blink); this isolates the quality of the screening stage from the downstream localisation.
- (iii) The count of channels the group requires, as a measure of montage cost, simplicity, and portability; a group that performs comparably with fewer channels is preferred because it is cheaper to acquire and easier to deploy.
- (iv) The fraction of non-blink epochs erroneously flagged and the overall proportion of epochs flagged; a low value indicates robustness to artefacts and avoids over-flagging that would dilute the Stage-B threshold estimate.
- (v) The agreement of the chosen group's behaviour between Raja and Cao2018; a group that is best on one dataset but poor on the other does not generalise.
- (vi) The stability of which channel (or channels) drives detection across sessions and subjects; an unstable best channel signals an overfitted, fragile selection.
- (vii) The consistency of the group's ranking under the median versus mean centre at Stage B; a group whose advantage

survives both estimators is more trustworthy than one that depends on a particular centre.

The configuration that best balances these criteria is designated the ideal configuration and is carried forward unchanged into Experiments 2 and 3.

b: Experiment 2 – Strategy comparison.

Purpose: benchmark the proposed pipeline, configured with the ideal channel group from Experiment 1, against competing detection algorithms. **Motivation:** a method is only convincing if it outperforms the established alternatives under identical preprocessing, epoching, and scoring. **Objective:** compare BLINKER-concat, MNE-annot, Proposed-Mean, and Proposed-Med at the event level on both datasets, reporting precision, recall, and F_1 with paired significance tests, and isolating both the estimator contrast (median versus mean) and the cross-dataset generalisation gap.

c: Experiment 3 – Epoch-duration stability.

Purpose: characterise how sensitive the proposed pipeline is to the choice of epoch length. **Motivation:** the 30 s primary epoch is a design choice, and a robust method should not depend critically on it. **Objective:** sweep the epoch duration (10/20/40/50/60/120 s around the 30 s reference) and quantify the resulting variation in event-level F_1 , testing each duration against the reference with Wilcoxon signed-rank tests.

F. EVALUATION AND STATISTICAL ANALYSIS

Detection was scored at the event level: a detected interval that overlapped an unmatched ground-truth interval was counted as a true positive, unmatched detections were counted as false positives, and unmatched ground-truth intervals were counted as false negatives. Event-level precision, recall, and F_1 were macro-averaged across sessions, and event-level F_1 was preferred over accuracy because blink events were sparse relative to non-blink samples. A detection was matched to a ground-truth interval at an intersection-over-union tolerance of 0.1. Methods were compared using paired two-tailed Wilcoxon signed-rank tests on session-level F_1 , with Bonferroni correction applied to the six comparisons among the four methods and to the six non-reference durations of the epoch sweep against the 30 s reference; rank-biserial effect sizes were also reported.

III. EXPERIMENTS AND RESULTS

A. EXPERIMENTAL SETUP

All experiments are evaluated on the driving corpora described in Section II-C. Unless otherwise stated, each continuous session is divided into non-overlapping 30-second epochs, and the evaluation is conducted at the event level using the overlap-based matching criterion defined in Section III-B.

a: Conditions.

Four conditions are compared throughout.

BLINKER-BLINK. The BLINKER algorithm is applied to the full concatenation of all valid epochs, treating the session as a single continuous signal and deriving a single threshold from the entire recording [8]. This condition represents the current practice of running a continuous-signal detector on epoch-structured data without any epoch-aware adaptation.

MNE-annot. The MNE-Python `annotate_amplitude` routine, which marks contiguous segments whose peak-to-peak amplitude exceeds a configurable threshold, used here with its default parameterisation. This baseline represents widespread preprocessing practice in the neuroimaging community and provides a reference point grounded in community tooling.

Proposed-Mean. The three-stage pipeline described in Section II, with the sample-level threshold at Stage B estimated from the arithmetic mean and standard deviation of samples pooled from suspicious epochs.

Proposed-Med. The three-stage pipeline with the sample-level threshold at Stage B estimated from the median and median absolute deviation (MAD). This is the primary proposed configuration.

b: Aggregation.

Every condition is summarised at its *best-channel-per-session* operating point: for each session the channel or channel group with the highest event-level F_1 is selected, and those per-session values are then averaged. The same rule is applied to all four conditions, so no condition is given a selection advantage the others do not receive.

B. EVALUATION METRICS

a: Event-level precision, recall, and F_1 .

A detected interval is counted as a true positive if it overlaps at least one unmatched ground-truth blink region; unmatched detections are false positives, and unmatched ground-truth regions are false negatives. Per-session precision, recall, and F_1 are computed from these event-level counts.

b: Macro versus micro F_1 .

We adopt **macro-averaged F_1** as the primary reporting metric. Macro F_1 computes precision, recall, and F_1 independently for each session and then reports the unweighted mean of the session-level F_1 scores. This gives every session equal weight regardless of how many blink events it contains. The goal of epoch-based blink detection in driving studies is to produce a detector that works reliably session by session, not one that merely accumulates favourable aggregate counts. Micro F_1 , which pools all true positives, false positives, and false negatives across sessions before computing a single ratio, is dominated by sessions with large blink counts and can mask consistent failure modes on sessions with atypically few blinks.

c: Wilcoxon signed-rank test.

Pairwise comparisons between conditions use the Wilcoxon signed-rank test on matched pairs of session-level F_1 scores. The test is preferred over a paired t -test because session-level F_1 distributions are bounded and often skewed. For comparisons in which the proposed method is hypothesised to outperform a baseline, a one-tailed test is used; stability analyses use two-tailed tests. Bonferroni correction is applied for the number of conditions compared within each experiment, and we report exact p -values alongside the matched-pairs rank-biserial correlation r as an effect-size estimate.

C. EXPERIMENT 1: EEG CHANNEL-SUBSET ANALYSIS

In the proposed pipeline, the Stage A screened epochs using a union rule over the channels provided to the pipeline, so changing the channel set, in theory, may also change which epochs reached Stage B. Restricting the available channels into pipeline, therefore changed the detector itself rather than merely changing the signals presented to a fixed detector. Therefore, the channel-subset analysis was performed by re-running the complete pipeline with each channel subset rather than evaluating a detector trained on a different channel set. But as a baseline, we first show how the detector performed when trained and tested on the full channel set, on a subset of the channels, and on a single channel.

1) Full-Montage Reference

First, we supply the complete 32 channel montage to the pipeline, which serves as a reference for the subsequent channel-subset analysis. This also interesting as we assume no prior knowledge of where blinks project, and the full montage is the most general configuration.

Against this full-montage reference, per-electrode detection quality was concentrated at the frontopolar pair rather than distributed across the frontal region, with each electrode scored within the 32-channel Proposed-Med run in which Stage A screened epochs using every electrode (Table 2, Table 3, and Figure 2). On Raja, Fp2 and Fp1 reached F_1 values of 0.8682 and 0.8632, respectively, before performance fell at the neighbouring frontal sites AF4 and AF3 to 0.8032 and 0.7376, with Fz, F3, and F4 at 0.6416, 0.4565, and 0.4523; the lowest value was 0.0175 at CP6.

On Cao2018, Fp1 and Fp2 reached F_1 values of 0.7968 and 0.7845, respectively, whereas the neighbouring frontal sites F7, F8, Fz, F3, and F4 were lower at 0.6028, 0.5971, 0.5970, 0.5951, and 0.5799; the lowest value was 0.0281 at Oz.

The central electrodes also showed a dataset-specific asymmetry: on Raja, C3 reached $F_1 = 0.6106$ while C4 reached $F_1 = 0.0185$, whereas on Cao2018 C3 and C4 were close at $F_1 = 0.3108$ and $F_1 = 0.3018$, respectively, so the pronounced C3–C4 asymmetry was specific to Raja.

2) Regional Channel Subsets

Following the preceding analysis, Experiment 1 tested whether channel subsets were sufficient as self-contained detectors by running the complete Stage A–Stage C Proposed-

TABLE 1. Raja EGI 128-channel (HydroCel GSN) to 10–20 scalp-location mapping, taken from the `egi_pair` block of `brain_region_raja.yaml`. Results are reported throughout by 10–20 location; this table gives the native EGI index for each one. The region column is the assignment used by the Experiment 1 run; `midline_or_outside` marks electrodes outside the curated left/right regions. Cao2018 is acquired directly in 10–20 nomenclature and needs no mapping.

10–20	EGI	Region
F7	E33	–
T7	E40	–
O2	E84	–
T8	E109	–
F8	E122	–
FC1	E13	central_left
C3	E26	central_left
FC5	E28	central_left
C4	E104	central_right
FC2	E112	central_right
FC6	E117	central_right
Fp1	E22	frontal_left
AF3	E23	frontal_left
F3	E24	frontal_left
AF4	E3	frontal_right
Fp2	E9	frontal_right
F4	E124	frontal_right
Fz	E11	midline_or_outside
Pz	E62	midline_or_outside
Oz	E75	midline_or_outside
Cz	E129	midline_or_outside
P7	E59	occipital_left
P3	E60	occipital_left
PO3	E67	occipital_left
O1	E70	occipital_left
PO4	E77	occipital_right
P4	E85	occipital_right
P8	E91	occipital_right
CP1	E37	parietal_left
CP5	E47	parietal_left
CP2	E87	parietal_right
CP6	E98	parietal_right

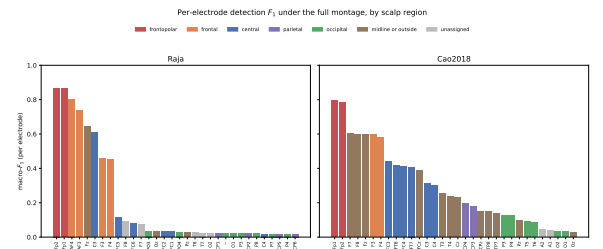


FIGURE 2. Per-electrode detection F_1 under the full montage, for every electrode, coloured by scalp region. Each bar is one electrode scored inside the 32-channel run, not a single-electrode pipeline.

Med pipeline separately on each subset in both Raja and Cao2018, using 30-second epochs and the median centre. Raja comprised 46 sessions and 20 channel subsets, including 14 anatomical subsets with the full montage and 6 single-electrode subsets, whereas Cao2018 comprised 58 sessions and 18 channel subsets, including 14 anatomical subsets with the full montage and 4 single-electrode subsets; in each corpus, 13 subsets were compared against the full montage.

Performance was aggregated by selecting the best channel for each session and then averaging over sessions. Compar-

TABLE 2. Experiment 1 channel-by-channel detection performance of Proposed-Med (median centre), organised by scalp region with no region-level aggregation. Each entry is the detection performance obtained on one electrode within the full-montage run, in which Stage A screened epochs using all 32 electrodes; these are not single-electrode results, which are reported in Table ?? . Precision, recall and F_1 are macro-averaged over all sessions (values read directly from the selection=all_channel rows of the Experiment 1 results). Electrodes are identified by their 10–20 scalp location on both corpora; for Raja the corresponding native EGI HydroCel index is given alongside it (full mapping in Table 1). Best F_1 per dataset in bold.

Region	Channel	EGI	Precision	Recall	F_1
<i>Raja (EGI 128, 46 sessions)</i>					
frontal_left	Fp1	E22	0.865	0.889	0.863
	AF3	E23	0.817	0.720	0.738
	F3	E24	0.642	0.397	0.456
frontal_right	Fp2	E9	0.863	0.897	0.868
	AF4	E3	0.869	0.777	0.803
	F4	E124	0.698	0.377	0.452
central_left	C3	E26	0.690	0.602	0.611
	FC5	E28	0.385	0.075	0.116
	FC1	E13	0.293	0.017	0.031
central_right	FC6	E117	0.374	0.051	0.081
	FC2	E112	0.276	0.018	0.031
	C4	E104	0.157	0.010	0.019
parietal_left	CP1	E37	0.150	0.012	0.022
	CP5	E47	0.122	0.010	0.018
	CP2	E87	0.173	0.011	0.020
parietal_right	CP6	E98	0.118	0.010	0.017
	PO3	E67	0.141	0.023	0.036
	O1	E70	0.123	0.012	0.021
occipital_left	P3	E60	0.124	0.012	0.021
	P7	E59	0.118	0.010	0.018
	PO4	E77	0.143	0.018	0.030
occipital_right	–	E83	0.117	0.013	0.021
	P8	E91	0.117	0.011	0.019
	P4	E85	0.114	0.010	0.018
midline_or_outside	Fz	E11	0.805	0.599	0.642
	Oz	E75	0.124	0.021	0.033
	Pz	E62	0.138	0.016	0.028
unassigned	F8	E122	0.296	0.060	0.092
	F7	E33	0.209	0.050	0.074
	T8	E109	0.170	0.015	0.027
	T7	E40	0.151	0.014	0.025
	O2	E84	0.117	0.013	0.022
<i>Cao2018 (10–20, 58 sessions)</i>					
frontal_left	Fp1	–	0.740	0.905	0.797
	F3	–	0.726	0.549	0.595
	Fp2	–	0.738	0.879	0.784
frontal_right	F4	–	0.717	0.532	0.580
	FC3	–	0.668	0.373	0.441
	FT7	–	0.621	0.336	0.408
central_left	C3	–	0.595	0.243	0.311
	FC4	–	0.643	0.344	0.416
	FC4	–	0.647	0.344	0.410
central_right	C4	–	0.579	0.234	0.302
	CP3	–	0.482	0.127	0.178
	CP4	–	0.492	0.140	0.196
parietal_left	P3	–	0.425	0.085	0.128
	T5	–	0.354	0.060	0.093
	O1	–	0.227	0.018	0.031
parietal_right	P4	–	0.429	0.083	0.127
	T6	–	0.351	0.052	0.083
	O2	–	0.256	0.019	0.033
occipital_left	F7	–	0.709	0.571	0.603
	F8	–	0.708	0.557	0.597
	Fz	–	0.727	0.556	0.597
occipital_right	FCz	–	0.633	0.323	0.389
	T3	–	0.524	0.188	0.256
	T4	–	0.535	0.167	0.236
midline_or_outside	Cz	–	0.511	0.174	0.231
	CPz	–	0.436	0.103	0.149
	TP8	–	0.448	0.098	0.149
unassigned	TP7	–	0.410	0.091	0.135
	Pz	–	0.389	0.062	0.098
	Oz	–	0.225	0.016	0.028
	A2	–	0.210	0.027	0.046
	A1	–	0.173	0.022	0.037

TABLE 3. Experiment 1 per-electrode detection performance collapsed to coarse scalp regions (Proposed-Med, median centre). Every electrode is scored inside the full-montage run, in which Stage A screens epochs using all 32 electrodes; these are therefore not single-electrode results, which are reported separately in Table ?? . Each region row averages the per-channel macro precision, recall and F_1 over the electrodes it contains; the frontopolar pair (Raja E22/E9, Cao2018 FP1/FP2) is reported separately from the remaining frontal sites. n is the number of electrodes in the region.

Dataset	Region	n	Precision	Recall	F_1
Raja	frontopolar	2	0.8636	0.8934	0.8657
	frontal	4	0.7564	0.5676	0.6124
	central	6	0.3626	0.1288	0.1480
	parietal	4	0.1409	0.0110	0.0194
	occipital	8	0.1247	0.0138	0.0230
	midline_or_outside	3	0.3557	0.2121	0.2342
Cao2018	unassigned	5	0.1885	0.0305	0.0480
	frontopolar	2	0.7389	0.8924	0.7906
	frontal	2	0.7217	0.5405	0.5875
	central	6	0.6257	0.3123	0.3813
	parietal	2	0.4870	0.1332	0.1870
	occipital	6	0.3404	0.0529	0.0824
	midline_or_outside	12	0.5213	0.2422	0.2890
	unassigned	2	0.1914	0.0244	0.0410

isons with the full montage used two-tailed Wilcoxon signed-rank tests with Bonferroni correction within each dataset, and matched-pairs rank-biserial correlation was reported as the effect size. Because Stages A–C were rerun using each subset alone, each subset constituted a self-contained detector, so this design measured whether a subset was sufficient on its own; it was not a leave-one-region-out analysis and did not establish that any region was necessary to the complete system. Raja electrodes were reported by their 10–20 scalp locations, with the corresponding native EGI indices given in Table 1.

At the anatomical-subset level, Proposed-Med retained performance comparable to the full montage only for frontal configurations, whereas central and more posterior subsets performed substantially worse (Table 4; Figure 3). The Frontal subset was statistically indistinguishable from the full montage on both corpora, with $F_1 = 0.8676$, a paired difference of -0.0149 , and $p = 1.000$ on Raja, and $F_1 = 0.7961$, a paired difference of -0.0106 , and $p = 1.000$ on Cao2018. After Bonferroni correction, Frontal, Frontal-Left and Frontal-Right did not differ significantly from the full montage on Raja, with $p = 1.000$, $p = 0.070$ and $p = 0.113$, respectively. On Cao2018, only Frontal and Frontal-Left remained non-significant, with $p = 1.000$ and $p = 0.123$, whereas Frontal-Right differed significantly from the full montage, with $p = 0.001$.

As expected, performance fell steeply when only central, parietal, occipital or posterior subsets were used. Raja yielded $F_1 = 0.5859$ for Central, $F_1 = 0.0264$ for Parietal, $F_1 = 0.0406$ for Occipital and $F_1 = 0.0426$ for Posterior, while Cao2018 yielded $F_1 = 0.4668$, $F_1 = 0.1668$, $F_1 = 0.1197$ and $F_1 = 0.1736$ for the corresponding subsets. All differences for these central, parietal, occipital and posterior subsets remained significant after correction, with large matched-

pairs rank-biserial effect sizes ($p < 0.001$ in every case; for Central, $r = -1.000$ on Raja and $r = -0.994$ on Cao2018).

In contrast to the frontal subsets, the non-frontal subsets failed through different precision–recall patterns in Raja and Cao2018, and their Stage-B thresholds did not follow a common ordering across the two corpora (Table 4). On Cao2018, the parietal, occipital, and posterior subsets retained precisions of 0.4944, 0.4361, and 0.5048, respectively, while recall fell to 0.1168, 0.0770, and 0.1194, with corresponding F_1 values of 0.1668, 0.1197, and 0.1736. On Raja, both measures fell for the same subsets: precision was 0.1952, 0.1638, and 0.1810, while recall was 0.0147, 0.0262, and 0.0275, with F_1 values of 0.0264, 0.0406, and 0.0426. The sample-level Stage-B blink-region threshold likewise differed markedly between subsets: on Cao2018, it decreased from 66.6 and 69.3 μV for the Fp1-only and Fp2-only subsets through 54.4 μV for the frontal subset, 33.6 μV for the central subset, and 29.9 μV for the parietal subset to 24.0 μV for the occipital subset. Raja did not show this ordering: the central-right subset had the highest threshold of any subset at 87.8 μV yet reached one of the lowest F_1 values, while the occipital threshold of 40.4 μV remained close to the full-montage value of 43.2 μV .

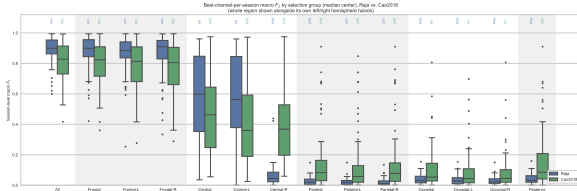


FIGURE 3. Session-level macro F_1 by channel-selection group (median centre), Raja versus Cao2018.

3) Single-Channel Operation

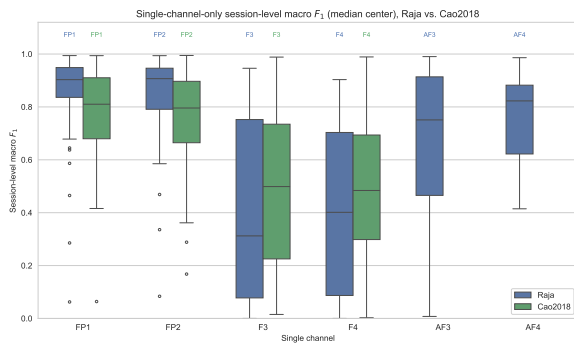


FIGURE 4. Session-level macro F_1 for the single-electrode selection groups, in which the complete pipeline was run on one electrode alone.

Consistent with the channel-subset results, the complete Proposed-Med pipeline retained most of the full-montage performance when operated on a single frontopolar electrode, whereas performance dropped sharply for single electrodes outside the frontopolar pair. The best single-electrode F_1 reached 0.8373 for Fp1 on Raja, and 0.7765 for Fp1 on

TABLE 4. Experiment 1 channel-subset performance of Proposed-Med (median centre) against the full-montage reference. Each subset is a self-contained detector: Stage A, Stage B and Stage C were all re-run on that channel subset alone. Precision, recall and F_1 are macro-averaged over sessions at the best-channel-per-session operating point; F_1 is the paired difference from the full montage, tested with a two-tailed Wilcoxon signed-rank test and Bonferroni-corrected over the 13 subsets compared within each dataset, with the matched-pairs rank-biserial correlation r as effect size. n is the number of electrodes in the subset; the frontal subset contains AF3 and AF4 on Raja but not on Cao2018, whose montage lacks them.

Subset	n	Precision	Recall	F_1	ΔF_1	p
<i>Raja (EGI 128, 46 sessions)</i>						
All (full montage)	32	0.8803	0.9014	0.8825	+0.0000	–
Frontal	6	0.8886	0.8741	0.8676	−0.0149	1.000
Frontal-L	3	0.8815	0.8569	0.8518	−0.0307	0.070
Frontal-R	3	0.8856	0.8475	0.8459	−0.0366	0.113
Central	6	0.6928	0.5642	0.5859	−0.2966	< 0.001
Central-L	3	0.6943	0.5467	0.5767	−0.3058	< 0.001
Central-R	3	0.4112	0.0429	0.0714	−0.8111	< 0.001
Parietal	4	0.1952	0.0147	0.0264	−0.8561	< 0.001
Parietal-L	2	0.1669	0.0119	0.0217	−0.8609	< 0.001
Parietal-R	2	0.2131	0.0119	0.0212	−0.8613	< 0.001
Occipital	8	0.1638	0.0262	0.0406	−0.8419	< 0.001
Occipital-L	4	0.1534	0.0211	0.0343	−0.8482	< 0.001
Occipital-R	4	0.1521	0.0188	0.0304	−0.8521	< 0.001
Posterior	12	0.1810	0.0275	0.0426	−0.8399	< 0.001
<i>Cao2018 (10–20, 58 sessions)</i>						
All (full montage)	32	0.7591	0.8909	0.8068	+0.0000	–
Frontal	4	0.7732	0.8578	0.7961	−0.0106	1.000
Frontal-L	2	0.7689	0.8423	0.7841	−0.0227	0.123
Frontal-R	2	0.7668	0.8132	0.7677	−0.0390	0.001
Central	6	0.6992	0.3894	0.4668	−0.3399	< 0.001
Central-L	3	0.6797	0.3358	0.4142	−0.3926	< 0.001
Central-R	3	0.6800	0.3376	0.4101	−0.3967	< 0.001
Parietal	2	0.4944	0.1168	0.1668	−0.6400	< 0.001
Parietal-L	1	0.4542	0.1033	0.1480	−0.6588	< 0.001
Parietal-R	1	0.4824	0.1151	0.1622	−0.6446	< 0.001
Occipital	6	0.4361	0.0770	0.1197	−0.6871	< 0.001
Occipital-L	3	0.4051	0.0630	0.0992	−0.7075	< 0.001
Occipital-R	3	0.4131	0.0622	0.0968	−0.7099	< 0.001
Posterior	8	0.5048	0.1194	0.1736	−0.6331	< 0.001

Cao2018. Outside the frontopolar pair, on Raja, F_1 was 0.7516 for AF4, 0.6759 for AF3, 0.3985 for F3, and 0.3998 for F4, while on Cao2018 it was 0.4930 for F3 and 0.4903 for F4 (Figure 4).

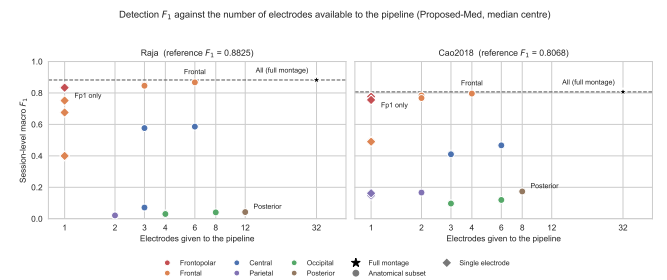


FIGURE 5. Detection F_1 against the number of electrodes given to the pipeline. Each point is one channel subset; the dashed line marks the full-montage reference.

TABLE 5. Best-channel selection frequencies pooled over the four conditions (best-channel-per-session). The 5 most frequently selected electrodes are listed per dataset, with the count and the fraction of all session $\times$ condition selections.

Dataset	Summary	Frequencies
Raja	Channels	Fp2 75/184 (0.408); Fp1 67/184 (0.364); AF4 14/184 (0.076); AF3 11/184 (0.060); Fz 4/184 (0.022)
Cao2018	Channels	Fp1 119/232 (0.513); Fp2 57/232 (0.246); F8 11/232 (0.047); Fz 11/232 (0.047); F7 9/232 (0.039)

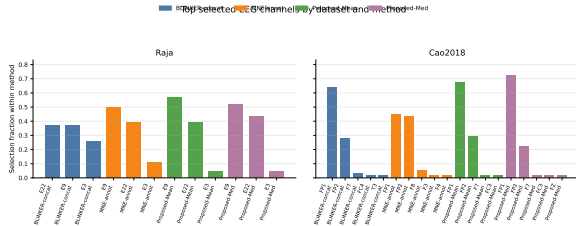


FIGURE 6. Best-channel selection frequency, pooled over the four conditions.

D. EXPERIMENT 2: STABILITY ACROSS EPOCH DURATIONS

Epoch duration is routinely chosen on paradigm-specific or theoretical grounds and is rarely optimised for the blink detector itself. In driving studies, epoch lengths ranging from a few seconds to several minutes appear in the literature. A robust pipeline should produce stable event-level F_1 across this range, because the underlying physiology of blinks does not change with the epoch grid imposed during analysis.

Proposed-Med is evaluated under epoch durations of 10, 20, 30, 40, 50, 60 and 120 seconds. For each duration the full pipeline is re-run from scratch, since epoch-level peak-to-peak statistics change with epoch length. Table 7 reports results and two-tailed Wilcoxon p -values relative to the 30-second reference. Across the epoch-duration sweep, Proposed-Med showed stable pooled performance and stable performance on Cao2018, whereas Raja differed significantly from the 30-second reference at several durations (Table 7, Figure 7). Pooled F_1 had a spread of 0.0039 across the seven durations, and no duration differed significantly from the 30-second reference after correction, with the smallest corrected p -value at 120 s ($p = 0.067$). Cao2018 likewise had no significant difference at any duration and showed a spread of 0.0046; the 120-second duration had the highest F_1 of 0.8112 but did not differ significantly from the 30-second reference ($p = 0.659$). Raja was not uniformly stable: relative to the 30-second reference, F_1 was 0.8793 at 50 s ($p < 0.001$), 0.8773 at 60 s ($p = 0.002$), and 0.8688 at 120 s ($p < 0.001$), with all three differences significant after correction. The 30-second reference had the highest Raja F_1 of 0.8825 among the seven durations.

E. EXPERIMENT 2: STRATEGY COMPARISON

In driving and other epoch-structured paradigms, it is common to apply a continuous-signal blink detector to the raw or

TABLE 6. Stability of the best-channel choice across the four conditions. Full agreement is the fraction of sessions where all four conditions select the same best channel; per-condition agreement is the mean fraction of the other three conditions selecting the same best channel.

Dataset	Sessions	Full agreement	Mean pairwise	BLINKER-concat	MNE-annot	Proposed-Med
Raja	46	9/46 (0.196)	0.489	0.442	0.326	0.513
Cao2018	58	6/58 (0.103)	0.408	0.397	0.218	0.513
Pooled	104	15/104 (0.144)	0.444	0.417	0.266	0.513

TABLE 7. Best-channel-per-session macro- F_1 of Proposed-Med across epoch durations. p -values (two-tailed Wilcoxon on session-level F_1 , Bonferroni-corrected over 6 non-reference durations) compare each duration against the 30 s reference. Bold marks the best duration within each block.

Dataset	Epoch duration	n	Macro F_1	p vs. 30 s
Raja	10 s	46	0.8771	1.000
	20 s	46	0.8821	1.000
	30 s	46	0.8825	reference
	40 s	46	0.8814	0.610
	50 s	46	0.8793	< 0.001
	60 s	46	0.8773	0.002
	120 s	46	0.8688	< 0.001
Cao2018	10 s	58	0.8077	1.000
	20 s	58	0.8076	1.000
	30 s	58	0.8068	reference
	40 s	58	0.8066	1.000
	50 s	58	0.8074	1.000
	60 s	58	0.8105	0.402
	120 s	58	0.8112	0.659
Pooled	10 s	104	0.8384	1.000
	20 s	104	0.8406	1.000
	30 s	104	0.8403	reference
	40 s	104	0.8397	0.933
	50 s	104	0.8392	0.523
	60 s	104	0.8401	1.000
	120 s	104	0.8367	0.067

minimally preprocessed recording and subsequently map the detected blinks back onto the epoch grid. Both BLINKER and the standard MNE pipeline follow this approach, in which channel-specific thresholds are derived from a signal pool containing both blink-heavy and blink-free epochs. The central premise of the proposed pipeline is that pooling these heterogeneous epochs dilutes the amplitude statistics used for threshold estimation, thereby distorting threshold placement and shifting the resulting precision–recall trade-off. Accordingly, we hypothesise that naive concatenation will produce systematically lower session-level F_1 scores than the epoch-aware pipeline because of a less favourable precision–recall trade-off. To evaluate this hypothesis, all four pipelines (i.e., BLINKER, MNE, Proposed-Med, and Proposed-Mean) are applied to every session using 30-second epochs and all available channels. All 32 channels are included to ensure a fair comparison with BLINKER, which, by default, processes all channels provided to it and requires the user to select a representative blink channel. For consistency across methods, all pipelines are therefore supplied with the same 32 channels, and the channel yielding the highest F_1 score is selected for subsequent analysis. The per-session macro- F_1 , precision,

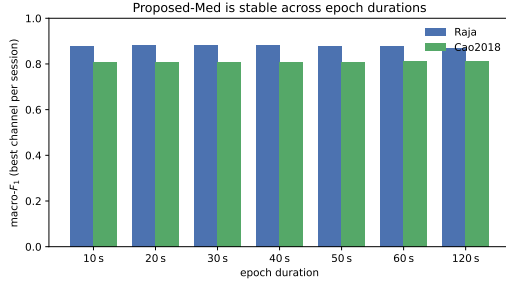


FIGURE 7. Macro-averaged F_1 of Proposed-Med across epoch durations.

and recall are summarised in Table 8.

In the strategy comparison, both proposed configurations achieved higher macro- F_1 scores than BLINKER and MNE on Raja, Cao2018, and the pooled evaluation (Table 8). Proposed-Med achieved macro- F_1 scores of 0.8825, 0.8068, and 0.8403, respectively, whereas Proposed-Mean achieved 0.8742, 0.8030, and 0.8345. The corresponding scores were 0.7654, 0.6940, and 0.7256 for BLINKER and 0.6664, 0.5627, and 0.6086 for MNE. This performance gap is reflected in the precision–recall profiles shown in Figure 8. BLINKER achieved the highest recall among all conditions on Raja, Cao2018, and the pooled evaluation, with values of 0.9518, 0.9715, and 0.9627, respectively, but also exhibited the lowest precision, at 0.6794, 0.5696, and 0.6182. Wilcoxon signed-rank tests on session-level F_1 , with Bonferroni correction across the six pairwise comparisons, showed significant differences between each proposed configuration and each baseline. Proposed-Med versus BLINKER yielded $p = 1.32 \times 10^{-11}$, and Proposed-Med versus MNE yielded $p = 6.51 \times 10^{-9}$. Similarly, Proposed-Mean versus BLINKER yielded $p = 4.97 \times 10^{-10}$, while Proposed-Mean versus MNE yielded $p = 9.94 \times 10^{-9}$. In contrast, neither within-pair comparison was significant: BLINKER versus MNE yielded $p = 0.812$, whereas Proposed-Mean versus Proposed-Med yielded $p = 1$. Furthermore, no channel group in either dataset produced an inversion in which a baseline matched or exceeded the performance of Proposed-Med (Table 9).

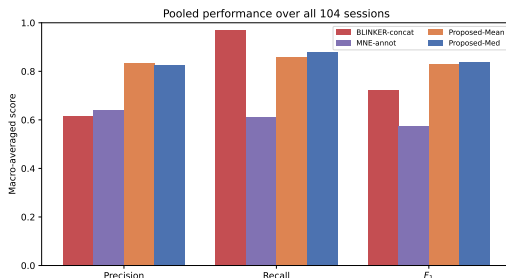


FIGURE 8. Macro-averaged precision, recall, and F_1 for the four conditions over all sessions.

1) Effect of the Threshold Estimator at Stage B

a: Motivation.

Stage B estimates the sample-level threshold from samples pooled across the suspicious epoch set $\mathcal{S}$. Because $\mathcal{S}$ is selected precisely because it contains high-amplitude excursions, the pooled distribution is not symmetric around a clean baseline: it is contaminated by the very peaks the threshold is meant to capture. The mean and standard deviation are sensitive to these extreme values; the median and MAD are not. The hypothesis is that the proposed-Med will outperform Proposed-Mean on precision—because the MAD-based threshold is less inflated by within-epoch peaks—while achieving comparable or superior recall.

Within the two proposed configurations, the Stage-B threshold estimators produced similar overall performance, although the direction of the metric changes did not support the stated precision hypothesis (Table 8). On Raja, Proposed-Med achieved precision, recall, and F_1 of 0.8803, 0.9014, and 0.8825, compared with 0.8840, 0.8876, and 0.8742 for Proposed-Mean. The corresponding changes were -0.0038 in precision, +0.0138 in recall, and +0.0083 in F_1 . On Cao2018, Proposed-Med likewise reduced precision by -0.0116 while increasing recall by +0.0190 and F_1 by +0.0037. Pooled performance followed the same pattern, with precision decreasing from 0.8209 to 0.8127, recall increasing from 0.8789 to 0.8956, and F_1 increasing from 0.8345 to 0.8403. However, the two estimators were statistically indistinguishable under the two-sided Bonferroni-corrected Wilcoxon test ($p = 1$). Proposed-Med exceeded Proposed-Mean in 46/104 sessions, was equal in 4, and was lower in 54, indicating a per-session outcome close to even.

2) Cross-Dataset Generalisation

a: Motivation.

A detector tuned and reported on a single corpus may not transfer to data collected under different acquisition conditions. Because the benchmark spans a closed corpus (Raja) and an open corpus (Cao2018), the generalisation gap can be quantified directly as the per-condition difference in macro- F_1 between the two datasets.

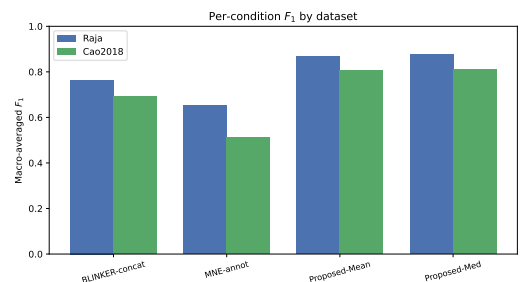


FIGURE 9. Macro-averaged F_1 per condition on Raja and Cao2018.

Across datasets, macro F_1 was consistently higher on Raja than on Cao2018 under every condition (Table 10; Figure 9). BLINKER-concat achieved 0.7654 on Raja and 0.6940 on Cao2018, corresponding to a cross-dataset gap of +0.0714.

TABLE 8. Strategy comparison on the Raja and Cao2018 driving-EEG corpora at 30 s epochs. Each condition is summarised at its best-channel-per-session operating point (for every session the single channel or frontal sub-montage with the highest F_1 is selected, then averaged over sessions); the same rule is applied to all four conditions. Macro-averaged precision, recall and F_1 are reported per dataset and pooled over all 104 sessions. Best F_1 per block in bold. By paired Wilcoxon signed-rank tests on session-level F_1 (Bonferroni-corrected over six pairs), Proposed-Med significantly exceeds BLINKER-concat ($p < 10^{-9}$) and MNE-annot ($p < 0.001$) but not Proposed-Mean ($p = 1.00$).

Dataset	Condition	Precision	Recall	F_1
Raja	BLINKER-concat	0.6794	0.9518	0.7654
	MNE-annot	0.6944	0.7063	0.6664
	Proposed-Mean	0.8840	0.8876	0.8742
	Proposed-Med	0.8803	0.9014	0.8825
Cao2018	BLINKER-concat	0.5696	0.9715	0.6940
	MNE-annot	0.6013	0.6168	0.5627
	Proposed-Mean	0.7707	0.8720	0.8030
	Proposed-Med	0.7591	0.8909	0.8068
Pooled	BLINKER-concat	0.6182	0.9627	0.7256
	MNE-annot	0.6425	0.6564	0.6086
	Proposed-Mean	0.8209	0.8789	0.8345
	Proposed-Med	0.8127	0.8956	0.8403

TABLE 9. Channel groups and datasets where a baseline algorithm equals or exceeds Proposed-Med in best-channel-per-session macro- F_1 .

Dataset	Selection	Baseline	BL- F_1	Prop- F_1	ΔF_1
No inversions — Proposed-Med leads on every channel group.					

TABLE 10. Cross-dataset generalisation gap (best-channel-per-session macro- F_1 , 30 s epochs). Gap = Raja – Cao2018.

Condition	Raja	Cao2018	Gap
BLINKER-concat	0.7654	0.6940	+0.0714
MNE-annot	0.6664	0.5627	+0.1037
Proposed-Mean	0.8742	0.8030	+0.0712
Proposed-Med	0.8825	0.8068	+0.0758

MNE-annot achieved 0.6664 and 0.5627, respectively, with the largest gap of +0.1037. Proposed-Mean achieved 0.8742 on Raja and 0.8030 on Cao2018, producing the smallest gap of +0.0712, whereas Proposed-Med achieved 0.8825 and 0.8068, respectively, with a gap of +0.0758. The two proposed configurations and BLINKER-concat therefore maintained similar, narrow differences between the two corpora, while MNE-annot showed the least consistent performance across datasets.

3) Operating Points

Across the pooled and corpus-specific analyses, the precision–recall operating points showed a consistent ordering among the proposed and baseline conditions (Figure 10 and Figure 11). In the pooled analysis, Proposed-Med and Proposed-Mean occupied the high-precision, high-recall region, with precision and recall of 0.8127 and 0.8956 for Proposed-Med and 0.8209 and 0.8789 for Proposed-Mean, respectively. BLINKER-concat occupied the high-recall, low-precision region, with recall of 0.9627 and precision of 0.6182, whereas MNE-annot was lower on both measures, with precision of 0.6425 and recall of 0.6564.

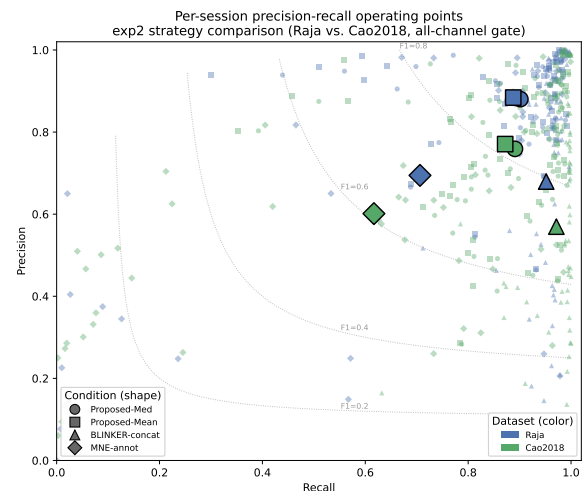


FIGURE 10. Per-session precision–recall operating points, Raja and Cao2018 shown separately.

The same ordering was observed within each corpus. On Raja, the proposed conditions had precision of 0.8840 and 0.8803 and recall of 0.8876 and 0.9014, whereas BLINKER-concat combined 0.9518 recall with 0.6794 precision and MNE-annot achieved 0.6944 precision and 0.7063 recall. On Cao2018, the corresponding pattern remained, with the proposed conditions retaining higher precision and recall than MNE-annot, while BLINKER-concat retained the highest recall and lower precision.

4) Blink-Count Fidelity

Event-level F_1 does not by itself establish that a detector can support blink-rate or PERCLOS estimation: that requires the predicted event count to track the true count. For each session the predicted count is $TP + FP$ and the true count is $TP + FN$.

Beyond the precision–recall operating points, blink-count fidelity also differed across conditions (Table 11; Figure 12). BLINKER-concat produced a mean predicted-to-true count

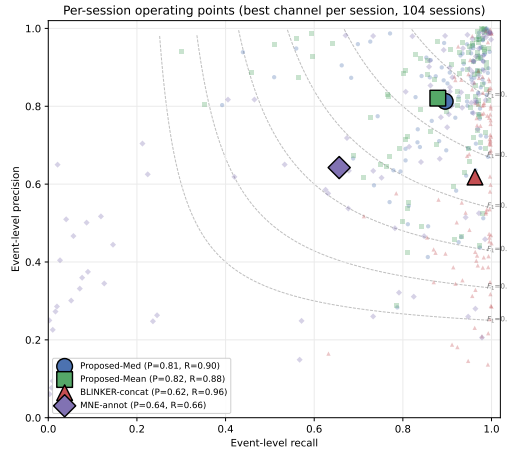


FIGURE 11. Per-session operating points pooled over both corpora, with condition means and iso- F_1 contours.

TABLE 11. Agreement between the predicted blink count (TP + FP) and the true count (TP + FN) per session, pooled over Raja and Cao2018 ($n = 104$) at the best-channel-per-session row. The mean count ratio measures systematic over- or under-counting (1.00 is ideal); Pearson r and Lin's concordance correlation coefficient (CCC) measure how closely the predicted count tracks the true count across sessions.

Condition	Mean count ratio	Pearson r	Lin's CCC
Proposed-Med	1.16	0.941	0.938
Proposed-Mean	1.12	0.918	0.917
BLINKER-concat	1.94	0.868	0.714
MNE-annot	1.04	0.629	0.617

ratio of 1.939, with Pearson $r = 0.868$ and Lin's concordance correlation coefficient of 0.714, indicating that it roughly doubled the true blink count despite retaining a positive association with the true counts. MNE-annot produced a count ratio close to one at 1.036, but its Pearson correlation and concordance were lower at $r = 0.629$ and 0.617, respectively. Its apparently accurate pooled total therefore resulted from offsetting over-counting and under-counting errors rather than consistent tracking of the true count. Proposed-Mean achieved a count ratio of 1.124, Pearson $r = 0.918$, and concordance of 0.917, whereas Proposed-Med achieved 1.157, $r = 0.941$, and 0.938, respectively. For Proposed-Med, Bland-Altman analysis gave a bias of +80.2 events, with $SD=303.1$ and 95% limits of agreement from -513.9 to $+674.4$, showing that it still over-counted on average.

5) Error-Structure Decomposition

Two detectors can reach the same F_1 by opposite routes: one floods the recording with false positives, the other misses blinks. Decomposing each condition's errors into false-positive and false-negative regimes distinguishes the two.

The corresponding error profiles differed across conditions (Table 12), particularly in the balance between false positives and false negatives. BLINKER-concat produced 627.9 false positives and 23.5 false negatives per session, giving an

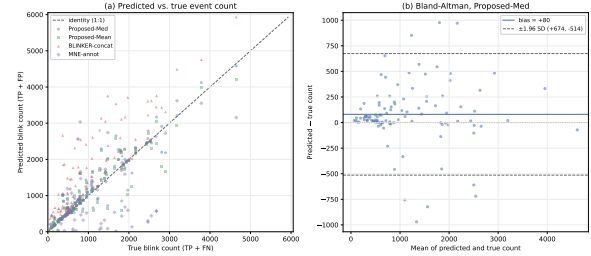


FIGURE 12. Predicted versus true blink count, and a Bland-Altman view for Proposed-Med.

TABLE 12. Error-structure decomposition by condition (best-channel-per-session). Mean false positives (FP) and false negatives (FN) per session, pooled over 104 sessions. The regime column records which error type dominates.

Condition	Mean FP/session	Mean FN/session	FP:FN	Regime
BLINKER-concat	627.9	23.5	26.675	FP-heavy
MNE-annot	276.7	451.5	0.613	FN-heavy
Proposed-Mean	191.3	158.7	1.206	FP-heavy
Proposed-Med	208.0	127.7	1.628	FP-heavy

FP:FN ratio of 26.675 and an overwhelmingly false-positive-heavy error profile. MNE-annot produced 276.7 false positives and 451.5 false negatives per session, with an FP:FN ratio of 0.613, indicating that its errors were false-negative-heavy. Both proposed configurations were closer to parity. Proposed-Mean produced 191.3 false positives and 158.7 false negatives per session, corresponding to an FP:FN ratio of 1.206, whereas Proposed-Med produced 208.0 false positives and 127.7 false negatives per session, with an FP:FN ratio of 1.628. Thus, both proposed configurations remained moderately false-positive-heavy, but Proposed-Med traded additional false positives for fewer false negatives than Proposed-Mean.

6) Per-Session Variability and Failure Analysis

TABLE 13. Best and worst Proposed-Med sessions and subject-level summary across 104 Raja+Cao2018 sessions (best-channel-per-session).

Scope	Dataset	Unit	n	Metric
Best session	Raja	S21/S34_20190122_044130_3	1	F_1
Worst session	Cao2018	S53/090925m	1	F_1
Median session	all	104 sessions	104	F_1
Best subject	Cao2018	S55	1	Mean F_1
Worst subject	Cao2018	S31	2	Mean F_1
Median subject	all	44 subjects	44	Mean F_1

Beyond these condition-specific error profiles, Proposed-Med showed substantial variability across sessions and subjects, and the lowest-scoring sessions did not share a uniform failure pattern (Table 13). Across 104 sessions, session-level F_1 ranged from 0.9944 for S21/S34_20190122_044130_3 in Raja to 0.4167 for S53/090925m in Cao2018, with a median of 0.8730 and an interquartile range of [0.7659, 0.9360]. Across 44 subjects, mean F_1 ranged from 0.9928

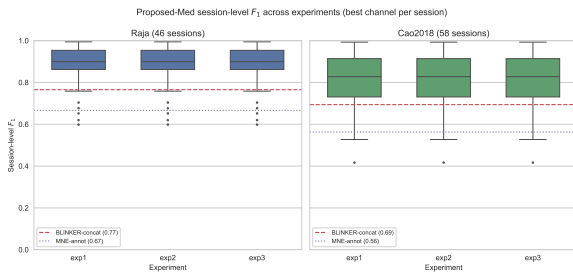
TABLE 14. The 5 lowest- F_1 Proposed-Med sessions per corpus (best channel per session). GT is the ground-truth blink count (TP + FN) and GT/med is GT relative to the dataset median, which separates genuine detector failure from recordings that simply carry an atypical number of blinks.

Dataset	Session	GT	TP	FP	FN	Regime	F_1	GT/med	Best channel
Raja	S16/S29_20190111_034326_3	1822	800	52	1022	FN-heavy	0.598	$3.5 \times$	Fp2
	S1/S01_20170519_043933_3	132	123	142	9	FP-heavy	0.620	$0.3 \times$	AF4
	S3/S03_20170605_033654_3	91	74	62	17	FP-heavy	0.652	$0.2 \times$	Fp1
	S18/S31_20190115_035853_2	455	313	157	142	FP-heavy	0.677	$0.9 \times$	Fp1
	S24/S38_20190129_035118_2	1117	625	34	492	FN-heavy	0.704	$2.1 \times$	Fp2
Cao2018	S53/090925m	367	289	731	78	FP-heavy	0.417	$0.3 \times$	Fp1
	S31/061020m	1476	579	142	897	FN-heavy	0.527	$1.1 \times$	Fp1
	S49/080527n	174	149	212	25	FP-heavy	0.557	$0.1 \times$	F7
	S22/090825n	382	372	479	10	FP-heavy	0.603	$0.3 \times$	Fp1
	S53/090918n	809	749	912	60	FP-heavy	0.606	$0.6 \times$	Fp1

for S55 to 0.5856 for S31, with a median subject mean of 0.8350. In Raja, 1 of 46 sessions had $F_1 < 0.6$ and 4 had $F_1 < 0.7$, whereas in Cao2018, 3 of 58 sessions had $F_1 < 0.6$ and 12 had $F_1 < 0.7$ (Table 14). The lowest-scoring sessions were not predominantly associated with anomalously high ground-truth blink counts: the bottom-5 median was 0.9x the Raja dataset median and 0.3x the Cao2018 dataset median. The bottom groups were also not uniformly false-negative-dominated, with FN-heavy cases in 2/5 Raja sessions and 1/5 Cao2018 sessions.

F. SUMMARY ACROSS EXPERIMENTS

Table 15 consolidates Proposed-Med performance across the three experiments, Figure 13 shows the corresponding session-level distributions, and Table 16 reports the paired comparison against the best competing method in every setting.

**FIGURE 13.** Session-level F_1 distributions of Proposed-Med across the three experiments.

Taken together, the three experiments showed that Proposed-Med consistently outperformed the best competing method, BLINKER-concat, on both Raja and Cao2018 (Table 15; Figure 13). The three experiments share a single primary configuration, the full montage at 30 s epochs with the median centre, and each reproduced it exactly: pooled macro F_1 was 0.8403 in all three, compared with 0.7256 for BLINKER-concat, a pooled advantage of +0.1147. The corpus-specific values were likewise identical across Experiments 1–3, at 0.8825 on Raja and 0.8068 on Cao2018. In the paired analyses, every bootstrap 95% confidence interval excluded zero, and all six one-sided Wilcoxon comparisons remained significant after Bonferroni correction (Table 16).

Thus, the advantage of Proposed-Med was consistent across experimental settings and corpora.

IV. DISCUSSION

Deriving the amplitude threshold from epoch-screened signal rather than from the whole concatenated recording provides a substantial and statistically significant improvement in blink-region detection across independent driving corpora. Proposed-Med achieved a pooled macro- F_1 of 0.8403, compared with 0.7256 for BLINKER-concat and 0.6086 for MNE-annot (Table 8). BLINKER-concat represented an established blink-specific method based on thresholding the concatenated recording [8], whereas MNE-annot represented a community preprocessing baseline related to automated rejection and standardized preprocessing procedures [26, 27]. The corresponding session-level differences were statistically significant for both BLINKER-concat ($p = 1.32 \times 10^{-11}$) and MNE-annot ($p = 6.51 \times 10^{-9}$), and neither baseline equalled or exceeded Proposed-Med in any channel group on either dataset. These findings indicate that estimating the threshold after excluding unsuitable epochs reduces the influence of recording-wide amplitude heterogeneity and yields more consistent region-level discrimination than applying a single threshold derived from the complete signal. This improvement is consistent with reduced contamination of the reference amplitude distribution by heterogeneous epoch composition. When blink-heavy and blink-free epochs were pooled across the concatenated recording, the large proportion of blink-free signal pulled the estimated threshold toward the baseline amplitude, causing the detector to fire too readily. The resulting error structure provides direct evidence for this mechanism: BLINKER-concat retained the highest pooled recall of 0.9627 but had a precision of 0.6182, with 627.9 false positives and 23.5 false negatives per session and an FP:FN ratio of 26.675. Its errors were therefore overwhelmingly false-positive-heavy rather than arising from missed blinks. This sensitivity is expected because ocular-artifact amplitude varies across subjects, electrodes, and recording sessions [6, 8, 11, 24]. A single threshold derived from the full recording consequently depends on the relative contribution of blink-rich and blink-poor periods, whereas epoch screening restricts threshold estimation to signal segments that better represent the local background. The observed precision gain therefore

TABLE 15. Proposed-Med detection performance across the three experiments on the Raja and Cao2018 driving-EEG corpora. Macro-averaged F_1 over all sessions (best-channel-per-session). The three rows are identical because the primary configuration of all three experiments is the same one: the full montage at 30-second epochs with the median centre. The rows are reported separately to show that each experiment reproduced it exactly. The best competing method is BLINKER-concat from the strategy comparison; Δ is the pooled Proposed-Med advantage over BLINKER-concat (pooled macro- F_1 0.7256).

Exp.	Description	PM F_1 (Raja)	PM F_1 (Cao2018)	Best competitor	Δ vs competitor
exp1	Channel subsets (full montage)	0.8825	0.8068	BLINKER-concat	+0.1147
exp2	Strategy comparison	0.8825	0.8068	BLINKER-concat	+0.1147
exp3	Epoch duration (30 s)	0.8825	0.8068	BLINKER-concat	+0.1147

TABLE 16. Paired Wilcoxon signed-rank tests of Proposed-Med versus the best competing method (BLINKER-concat from the strategy comparison) on session-level F_1 , for each experiment and dataset. ΔF_1 is the mean Proposed-Med advantage; p is Bonferroni-corrected over the 6 comparisons (one-sided, Proposed-Med greater); r is the rank-biserial effect size; the 95% CI is a 10,000-sample bootstrap on ΔF_1 .

Exp.	Dataset	ΔF_1	W	p_{Bonf}	r	95% CI	n
exp1	Raja	+0.1171	909	$< 10^{-5}$	0.159	[+0.0663, +0.1699]	46
exp1	Cao2018	+0.1127	1613	$< 10^{-8}$	0.057	[+0.0861, +0.1394]	58
exp2	Raja	+0.1171	909	$< 10^{-5}$	0.159	[+0.0655, +0.1699]	46
exp2	Cao2018	+0.1127	1613	$< 10^{-8}$	0.057	[+0.0860, +0.1398]	58
exp3	Raja	+0.1171	909	$< 10^{-5}$	0.159	[+0.0646, +0.1691]	46
exp3	Cao2018	+0.1127	1613	$< 10^{-8}$	0.057	[+0.0856, +0.1395]	58

reflects improved calibration to session-specific amplitude structure rather than reduced sensitivity to true blinks. The pipeline's advantage therefore arises from screening epochs before threshold estimation [28] rather than from whether the retained amplitudes are centred by the median or the mean. The median/MAD centre was selected to reduce the influence of within-epoch peaks that remained in the suspicious-epoch pool, consistent with the general robustness of median-based estimators to outlying observations [29, 30]. This rationale was not supported by a measurable performance gain in the present benchmark. Relative to Proposed-Mean, Proposed-Med reduced pooled precision by 0.0081 while increasing recall by 0.0167, yielding only a 0.0057 increase in F_1 . The two estimators were statistically indistinguishable after correction ($p = 1$), and Proposed-Med exceeded Proposed-Mean in 46/104 sessions, with 4 equal and 54 lower. Thus, the median centre did not provide the hypothesised precision benefit and won only about half of the sessions. Its resistance to contaminated amplitude samples makes it a defensible default on statistical grounds, but the present results do not support it as a source of measurable improvement.

A. SPATIAL REQUIREMENTS

The spatial requirements of the pipeline were limited because a single frontopolar electrode recovered essentially all of the full-montage performance in both corpora. Scored within the full montage, Fp2 achieved a macro- F_1 of 0.8682 on Raja against the full-montage reference of 0.8825, and Fp1 achieved 0.7968 on Cao2018 against 0.8068; running the complete pipeline on that electrode alone, with no montage behind it, still reached 0.8337 and 0.7765 respectively. The optimum therefore localised consistently to Fp1/Fp2 despite differences in montage and acquisition hardware, while performance declined at electrodes farther from the eyes. This

spatial consistency is compatible with the strong projection of blink activity to the prefrontal scalp and indicates that additional channels contributed little to blink-region detection once a frontopolar signal was available. The result also agrees with prior single-channel and forehead-EEG studies that detected blinks or derived fatigue-related measures from prefrontal recordings [10, 12, 15, 16, 21, 23, 31–33]. A dense montage is therefore not required for this task, which supports implementations based on a single dry or embedded forehead electrode in wearable and in-vehicle systems where setup time, comfort, and sensor count constrain deployment. Within this reduced spatial requirement, channel selection was stable at the level of scalp region but not at the level of a single electrode. Fp1 and Fp2 accounted for the large majority of selections in both corpora: Raja favoured Fp2 in 75/184 selections and Fp1 in 67/184, whereas Cao2018 favoured Fp1 in 119/232 and Fp2 in 57/232. However, the four conditions selected the same channel in only 9/46 Raja sessions and 6/58 Cao2018 sessions, with pooled full agreement of 15/104. This apparent tension indicates that the detector consistently localised blink information to the frontopolar region while the electrode carrying the strongest signal varied across conditions and sessions. Such variation is consistent with lateralised ocular projections and inter-subject differences in blink topography, electrode placement, and signal amplitude rather than with instability of the underlying detection principle [24, 34]. The practical implication is that deployment should retain both frontopolar sites and select between them adaptively, rather than fixing acquisition or detection to Fp1 or Fp2 alone.

B. GENERALISATION AND ROBUSTNESS

Beyond this spatial consideration, cross-dataset generalisation was configuration-dependent: all conditions achieved

higher macro- F_1 on Raja than on Cao2018, but the magnitude of this difference varied by configuration. Proposed-Mean and Proposed-Med showed gaps of +0.0712 and +0.0758, respectively, which were close to the +0.0714 gap obtained by BLINKER-concat. By contrast, MNE-annot showed the largest gap at +0.1037 and was therefore the least consistent across the two corpora. This pattern indicates that the lower scores on Cao2018 are more plausibly attributable to differences between the corpora than to overfitting specific to the proposed thresholding procedure. Cao2018 was collected as an open multichannel sustained-attention driving dataset [25], and differences in acquisition hardware, electrode montage, and recording conditions can alter blink amplitude and morphology across datasets. The proposed configurations nevertheless retained a cross-dataset gap comparable to that of the established blink-specific baseline [8]. Their advantage was therefore not confined to Raja, although absolute performance remained dependent on the recording corpus. Taken together, the epoch-duration sweep indicated that Proposed-Med tolerated changes in epoch duration when performance was pooled across Raja and Cao2018, with no tested duration differing significantly from the 30-second reference and a pooled F_1 spread of 0.0039 across the seven durations (Table 7; Figure 7). This tolerance was important because epoch duration in driving studies was determined by the experimental paradigm rather than by the blink detector, so the thresholding procedure needed to accommodate different segmentation choices. The exception was corpus-specific: on Raja, F_1 declined from 0.8825 at the 30-second reference to 0.8793 at 50 seconds ($p < 0.001$), 0.8773 at 60 seconds ($p = 0.002$), and 0.8688 at 120 seconds ($p < 0.001$), with the largest decline occurring at 120 seconds. One plausible interpretation was that a longer epoch was more likely to contain both blink and blink-free signal, making its summary amplitude less representative of either state and thereby diluting the screening stage; this would correspond to the same form of dilution that the pipeline was designed to avoid by estimating the threshold from screened epochs, although the present data did not establish this mechanism. Because Raja also declined significantly at 50 and 60 seconds, the safe operating range indicated by these data was narrower than the full tested range, and the 30-second setting was the setting supported on both corpora.

C. DOWNSTREAM USABILITY

For downstream usability, event-level F_1 did not by itself establish suitability for blink-rate or PERCLOS-style measures, because these applications require predicted counts to vary with the true counts across recordings [2, 4, 35, 36]. Proposed-Med provided the closest count fidelity, with a mean predicted-to-true ratio of 1.157, Pearson $r = 0.941$, and Lin CCC = 0.938. BLINKER-concat instead produced approximately twice the true count, with a ratio of 1.939, despite retaining a relatively high correlation of $r = 0.868$; this pattern indicates systematic inflation rather than accurate absolute measurement. MNE-annot showed the converse limita-

tion: its ratio of 1.036 was near unity, but its lower correlation of $r = 0.629$ and CCC = 0.617 indicate that over-counting and under-counting offset in the pooled mean rather than that session-level counts were faithfully tracked. Proposed-Med therefore better preserves between-session variation needed for downstream ocular indices. However, its positive bias of +80.2 events and ratio above unity show that it still over-counted, so absolute blink rates and derived thresholds should be interpreted with care. Consistent with this downstream focus, the error profiles placed the two baselines at opposite operating-point extremes, whereas the proposed methods provided a closer balance between false positives and false negatives. BLINKER-concat was strongly false-positive dominated, with an FP:FN ratio of 26.675, whereas MNE-annot was false-negative dominated, with a ratio of 0.613. By contrast, Proposed-Mean and Proposed-Med produced ratios of 1.206 and 1.628, respectively, indicating substantially closer correspondence between over-detection and missed events. This balance matters because methods with similar F_1 can impose different downstream costs: excessive false positives inflate estimated blink counts and increase the burden of rejecting spurious regions, whereas excessive false negatives suppress genuine events and bias analyses that depend on blink occurrence or timing. An operating point closer to parity therefore provides a more usable compromise when neither error type can be treated as negligible. Precision-recall characterisation is the appropriate framework for this comparison because blink detection is an imbalanced event-detection problem in which the positive class is sparse relative to continuous background signal [37]. Event-level evaluation is also preferable to sample-wise agreement because it assesses whether temporally localised detections correspond to distinct underlying events [38]. Beyond these aggregate error profiles, performance variability across sessions was not primarily explained by event density but was more consistent with differences in signal quality and the sensitivity of sparse-blink sessions to individual detection errors. Performance varied substantially across the 104 sessions, from an F_1 of 0.9944 to 0.4167, with a median of 0.8730 and an IQR of [0.7659, 0.9360]. Subject-level means likewise ranged from 0.9928 to 0.5856, with a median of 0.8350. However, the lowest-scoring sessions were not those with anomalously high blink counts. The bottom five sessions had median ground-truth counts of 0.9x the dataset median on Raja and only 0.3x on Cao2018, despite dataset medians of 526 and 1314, respectively. Their errors were also not uniformly dominated by missed events, because only 2/5 Raja sessions and 1/5 Cao2018 sessions were FN-heavy. This pattern argues against event density as the primary failure mechanism. In sparse-blink sessions, a small number of detection errors imposes a larger penalty on the per-session score. Ambiguity in blink definition and heterogeneity in artifact morphology may further increase this sensitivity across recordings [1, 39].

D. POSITIONING AGAINST PRIOR WORK

The proposed detector is positioned qualitatively rather than as numerically superior to earlier blink detectors because direct numerical ranking would be misleading given differences in datasets, blink definitions, sampling rates, and scoring procedures, as summarised in Table 17. In particular, matching criteria ranged from sample-level classification to event-level detection, so apparently similar performance measures did not necessarily represent the same detection requirement. This heterogeneity applies to threshold-based and unsupervised approaches alike [8, 9, 13, 15–17, 22]. Its performance was competitive with interpretable single-channel threshold detectors while being assessed under a strict event-level overlap criterion on both independent driving corpora. This evaluation required each predicted region to correspond adequately to a reference blink event rather than rewarding isolated correctly labelled samples. The detector also avoided the dataset-specific threshold tuning commonly used by comparable approaches, which indicates that its relative performance was obtained under a more transferable operating procedure rather than through adaptation to a particular recording set.

TABLE 17. Qualitative positioning of the present detector against prior single-channel and threshold-based blink/EOG detectors cited in this work. Because the prior studies use different datasets, blink definitions, sampling rates and especially different matching criteria (sample-level versus event-level), their reported metrics are not directly comparable to an event-level F_1 ; such cells are marked “n/r” (not reported in comparable form) rather than populated with non-comparable numbers. Only the final row reports quantitative detection figures, recomputed at the best-channel-per-session event-level operating point (macro- F_1 on Raja / Cao2018). The comparison is therefore qualitative by design.

Method	Signal channel	/	Evaluation data	Matching criterion	Reported metric
Chang et al. [15]	et	Single pre-frontal EEG	Own EEG	n/r	n/r
Tran et al. [16]		Single/few-channel EEG	Own EEG	n/r	n/r
Zhang et al. [12]	et	Frontal EEG (real-time)	Own EEG	n/r	n/r
Wang et al. [21]	et	Single-channel EEG	Own EEG	n/r	n/r
BLINKER [8]		Single channel (EEG/EOG)	Multiple corpora	n/r	n/r
Cao et al. [22]		Multi-channel EEG	Own EEG	n/r	n/r
Agarwal & Sivakumar [9]		Single-channel EEG	Own EEG	n/r	n/r
Valderrama et al. [23]	et	Single-channel EOG/EEG	Own recordings	n/r	n/r
Guttmann-Flury et al. [24]	et	Multi-channel EEG	Own EEG	n/r	n/r
This work (Proposed-Med)		Single frontopolar EEG	Raja + Cao2018 driving	Event-level overlap (IoU 0.1)	macro- F_1 0.88 / 0.81

E. LIMITATIONS AND FUTURE WORK

Beyond the constraints on direct comparison with earlier detectors, this study had several limitations. The benchmark was restricted to both driving corpora, and generalisation to other experimental paradigms or clinical populations was therefore not tested. Event detection was evaluated using a single matching tolerance, although the reported scores would change under a stricter overlap criterion; absolute F_1 values should consequently not be compared across studies without accounting for differences in event-matching rules [38]. Recall was not stratified by blink duration, so performance on prolonged eye closures, which are particularly relevant to drowsiness assessment, was not established. Ground truth was based on expert annotation of frontopolar EEG and therefore inherited the annotator’s operational definition of blink onset and offset, which remains non-trivial because blink boundaries depend on the signal representation and adopted criteria [1]. Finally, the proposed method is a thresholding detector and was not evaluated as superior to learned detectors such as transformer-, vision-, and deep-learning-based approaches [40–42]. Its intended advantages are instead methodological simplicity, interpretability, and the absence of per-dataset training. Addressing these limitations requires future work on dedicated long-closure detection, broader benchmarking, and integration with learned detectors. A dedicated long-closure module should first score normal blinks and prolonged closures separately. This distinction is necessary because the events most directly associated with drowsiness are sustained eye closures rather than ordinary blinks, and aggregating both categories under a single detection criterion may obscure the signal of greatest operational relevance [3, 5]. The benchmark should also be extended beyond driving data to other epoch-structured paradigms, including settings in which task blocks, trials, or recording segments provide natural units for estimating a representative signal distribution. Such evaluation would determine whether epoch screening remains useful when blink morphology, cognitive demands, and recording conditions differ from those in sustained driving. A further direction is to combine the screening stage with learned blink and drowsiness detectors rather than treating the approaches as alternatives [43–45]. In that configuration, epoch screening would serve as a computationally inexpensive front end that limits implausible candidate regions before a learned model performs the more specific classification.

V. CONCLUSION

The study addressed blink-region detection in epoch-structured EEG, for which concatenated-signal methods pooled blink-containing and blink-free epochs when estimating detection thresholds. The proposed pipeline screened epochs by peak-to-peak amplitude, estimated a sample-level threshold from suspicious epochs using either the mean or median centre with MAD-based scaling, applied the threshold to extract blink regions, and mapped the detections to epoch-local coordinates. At 30 s, the proposed vari-

ants improved pooled macro-averaged F_1 from 0.7256 for BLINKER-concat and 0.6086 for MNE-annot to 0.8345 for Proposed-Mean and 0.8403 for Proposed-Med. This improvement chiefly reflected higher precision, with pooled values of 0.8209 and 0.8127 compared with 0.6182 and 0.6425 for the baselines, while recall remained 0.8789 and 0.8956. Proposed-Mean and Proposed-Med were statistically indistinguishable on session-level F_1 ($p = 1$), indicating that the advantage arose from epoch-aware screening rather than from the choice of centre.

The remaining experiments showed that the pipeline retained its performance under reduced channel coverage and changes in epoch duration. A single frontopolar electrode recovered most of the full-montage performance: running the complete pipeline on one electrode alone gave F_1 values of 0.8337 for Fp2 against the full-montage reference of 0.8825 in Raja, and 0.7765 for Fp1 against 0.8068 in Cao2018. Pooled macro F_1 also remained stable from 10 s to 120 s, with a spread of 0.0039 and values ranging from 0.8367 to 0.8406, although on Raja alone performance declined significantly at 50, 60 and 120 s. The predicted blink count tracked the true count closely for both proposed variants, with Pearson $r = 0.918$ and Lin's CCC = 0.917 for Proposed-Mean and Pearson $r = 0.941$ and Lin's CCC = 0.938 for Proposed-Med. This agreement was sufficient to support blink-rate-style measures, although Proposed-Med retained a positive bias of +80.2 events and 95% limits of $[-513.9, +674.4]$.

The conclusions were limited by evaluation on the Raja and Cao2018 driving corpora, the use of a single event-matching tolerance, and the absence of stratification by blink duration. Future work will therefore assess the pipeline on additional recording settings and matching criteria, examine performance across blink-duration groups, and develop the planned dedicated long-closure module.

REFERENCES

- [1] M. Nyström, R. Andersson, D. Niehorster, R. Hessels, and I. Hooze, "What is a blink? classifying and characterizing blinks in eye openness signals," *Behavior Research Methods*, 2024, csv_key=28ULZ92A.
- [2] E. Alyan, "Blink-related eeg activity measures cognitive load during proactive and reactive driving," *Scientific Reports*, 2023, csv_key=alyan2023blink.
- [3] Y. Tian and J. Cao, "Fatigue driving detection based on electrooculography: a review," *Eurasip Journal on Image and Video Processing*, 2021, csv_key=GNNI8WH2.
- [4] P. Ren, X. Ma, W. Lai, M. Zhang, S. Liu, Y. Wang, M. Li, D. Ma, Y. Dong, Y. He, and X. Xu, "Comparison of the use of blink rate and blink rate variability for mental state recognition," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2019, csv_key=TUNW8X4S.
- [5] M. Shahbakhti, M. Beiramvand, E. Nasiri, S. Far, W. Chen, J. Sole-Casals, M. Wierzchon, A. Broniec-Wojcik, P. Augustyniak, and V. Marozas, "Fusion of eeg and eye blink analysis for detection of driver fatigue," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2023, csv_key=WL52M5AF.
- [6] R. J. Croft and R. J. Barry, "Removal of ocular artifact from the eeg: a review," *Neurophysiologie Clinique/Clinical Neurophysiology*, 2000, csv_key=croft2000removal.
- [7] S. Akuthota, K. Kumar, and R. Janapati, "Artifacts removal techniques in eeg data for bci applications: A survey," *Computational Intelligence and Deep Learning Methods for Neuro-rehabilitation Applications*, 2023, csv_key=7FIPNU96.
- [8] K. K. Kleifges, N. Bigdely-Shamlo, S. E. Kerick, and K. A. Robbins, "Blinker: Automated extraction of ocular indices from eeg enabling large-scale analysis," *Frontiers in Neuroscience*, 2017, csv_key=kleifges2017blinker.
- [9] M. Agarwal and R. Sivakumar, "Blink: A fully automated unsupervised algorithm for eye-blink detection in eeg signals," 2019, csv_key=C2X825UQ.
- [10] L.-W. Ko, O. Komarov, W.-K. Lai, W.-G. Liang, and T.-P. Jung, "Eyeblink recognition improves fatigue prediction from single-channel forehead eeg in a realistic sustained attention task," *Journal of Neural Engineering*, 2020, csv_key=Y3Y2KSSK.
- [11] A. Klein and W. Skrandies, "A reliable statistical method to detect eyeblink-artefacts from electroencephalogram data only," *Brain Topography*, 2013, csv_key=klein2013reliable.
- [12] Y. Zhang, X. Zheng, W. Xu, and H. Liu, "Rt-blink: A method toward real-time blink detection from single frontal eeg signal," *IEEE Sensors Journal*, 2023, csv_key=JFTJCTE7.
- [13] M. Rashida and M. Habib, "Quantitative eeg features and machine learning classifiers for eye-blink artifact detection: A comparative study," *Neuroscience Informatics*, 2023, csv_key=G28PH5JQ.
- [14] Y. Jiang, D. Wu, J. Cao, L. Jiang, S. Zhang, and D. Wang, "Eyeblink detection algorithm based on joint optimization of vme and morphological feature extraction," *IEEE Sensors Journal*, 2023, csv_key=QAZH DUHB.
- [15] W.-D. Chang, H.-S. Cha, K. Kim, and C.-H. Im, "Detection of eye blink artifacts from single prefrontal channel electroencephalogram," *Computer Methods and Programs in Biomedicine*, 2016, csv_key=EXMLDRXM.
- [16] D.-K. Tran, T.-H. Nguyen, and T.-N. Nguyen, "Detection of eeg-based eye-blinks using a thresholding algorithm," *European Journal of Engineering and Technology Research*, 2021, csv_key=tran2021detection.
- [17] L. Jefri, F. Rahman, N. Malik, and F. Isa, "Eye blink region detection in eeg using amplitude thresholding," 2022, csv_key=ECTDKUZE.
- [18] H. Nolan, R. Whelan, and R. Reilly, "Faster: Fully automated statistical thresholding for eeg artifact rejection," *Journal of Neuroscience Methods*, vol.

- 192, no. 1, pp. 152–162, 2010. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0165027010003894>
- [19] R. Ghosh, S. Phadikar, N. Deb, N. Sinha, P. Das, and E. Ghaderpour, “Automatic eyeblink and muscular artifact detection and removal from eeg signals using k-nearest neighbor classifier and long short-term memory networks,” *IEEE Sensors Journal*, 2023, csv_key=D62QMUQY.
 - [20] M. Wang, J. Wang, X. Cui, T. Wang, T. Jiang, F. Gao, and J. Cao, “Multidimensional feature optimization based eye blink detection under epileptiform discharges,” *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2022, csv_key=ARYZLKPY.
 - [21] G. Wang, “Sliding window higher-order cumulants for detection of eye blink artifact from short segments of single-channel eeg,” *PeerJ Computer Science*, 2025, csv_key=wang2025sliding.
 - [22] J. Cao, L. Chen, D. Hu, F. Dong, T. Jiang, W. Gao, and F. Gao, “Unsupervised eye blink artifact detection from eeg with gaussian mixture model,” *IEEE Journal of Biomedical and Health Informatics*, 2021, csv_key=H22KDJVS.
 - [23] J. Valderrama, A. De La Torre, and B. Van Dun, “An automatic algorithm for blink-artifact suppression based on iterative template matching: Application to single channel recording of cortical auditory evoked potentials,” *Journal of Neural Engineering*, 2018, csv_key=HK732W3F.
 - [24] E. Guttmann-Flury, X. Sheng, D. Zhang, and X. Zhu, “A new algorithm for blink correction adaptive to inter- and intra-subject variability,” *Computers in Biology and Medicine*, 2019, csv_key=guttmann2019new.
 - [25] Z. Cao, C.-H. Chuang, J.-K. King, and C.-T. Lin, “Multi-channel eeg recordings during a sustained-attention driving task,” *Scientific Data*, 2019, csv_key=cao2019multichannel.
 - [26] M. Jas, D. A. Engemann, Y. Bekhti, F. Raimondo, and A. Gramfort, “Autoreject: Automated artifact rejection for MEG and EEG data,” *NeuroImage*, vol. 159, pp. 417–429, 2017.
 - [27] N. Bigdely-Shamlo, T. Mullen, C. Kothe, K.-M. Su, and K. A. Robbins, “The PREP pipeline: Standardized preprocessing for large-scale EEG analysis,” *Frontiers in Neuroinformatics*, vol. 9, p. 16, 2015.
 - [28] D.-K. Tran, T.-H. Nguyen, and B.-V. Ngo, “Amplitude thresholding of eeg signals for eye blink and saccade detection,” 2021, pp. 268–273.
 - [29] C. Leys, C. Ley, O. Klein, P. Bernard, and L. Licata, “Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median,” *Journal of Experimental Social Psychology*, vol. 49, no. 4, pp. 764–766, 2013.
 - [30] P. J. Rousseeuw and C. Croux, “Alternatives to the median absolute deviation,” *Journal of the American Statistical Association*, vol. 88, no. 424, pp. 1273–1283, 1993.
 - [31] M. Shahbakhti, M. Beiramvand, I. Rejer, P. Augustyniak, A. Broniec-Wojcik, M. Wierzchon, and V. Marozas, “Simultaneous eye blink characterization and elimination from low-channel prefrontal eeg signals enhances driver drowsiness detection,” *IEEE Journal of Biomedical and Health Informatics*, 2022, csv_key=KPMMW2C5.
 - [32] Y. Aoki and T. Fukami, “Drowsiness detection based on eeg and blink information measured with a single electrode,” *International Journal of Innovative Computing, Information and Control*, vol. 19, no. 1, pp. 229–238, 2023.
 - [33] R. López-Ahumada, R. Jiménez-Naharro, and F. Gómez-Bravo, “A hardware-based configurable algorithm for eye blink signal detection using a single-channel bci headset,” *Sensors*, vol. 23, no. 11, 2023.
 - [34] M. Plöchl, J. P. Ossandón, and P. König, “Combining EEG and eye tracking: Identification, characterization, and correction of eye movement artifacts in electroencephalographic data,” *Frontiers in Human Neuroscience*, vol. 6, p. 278, 2012.
 - [35] L. Dai, Y. Li, and M. Zhang, “Detection of operator fatigue in the main control room of a nuclear power plant based on eye blink rate, perclos and mouse velocity,” *Applied Sciences (Switzerland)*, 2023, csv_key=5J3QBV3S.
 - [36] M. Zulkarnanie, K. Shanmugam, N. Badruddin, and M. Saad, “Enhancements to perclos algorithm for determining eye closures,” 2022, csv_key=6C9D6CVR.
 - [37] T. Saito and M. Rehmsmeier, “The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets,” *PLOS ONE*, 2015, csv_key=saito2015precision.
 - [38] R. Salles, “Softed: Metrics for soft evaluation of time series event detection,” *Computers & Industrial Engineering*, 2024, csv_key=salles2024softed.
 - [39] P. Arpaia, M. De Luca, L. Di Marino, D. Duran, L. Gargiulo, P. Lanteri, N. Moccaldi, M. Nalin, M. Picciafuoco, R. Robbio, and E. Visani, “A systematic review of techniques for artifact detection and artifact category identification in electroencephalography from wearable devices,” *Sensors*, 2025, csv_key=LKNZDL8L.
 - [40] Á. Fodor, K. Fenech, and A. Lőrincz, “Blinklinmult: Transformer-based eye blink detection,” *Journal of Imaging*, 2023, csv_key=8LGNBJAX.
 - [41] J. Hong, J. Shin, J. Choi, and M. Ko, “Robust eye blink detection using dual embedding video vision transformer,” 2024, csv_key=7IZFRWNY.
 - [42] A. Egambaram and N. Badruddin, “An investigation to detect driver drowsiness from eye blink artifacts using deep learning models,” 2022, pp. 23–29.
 - [43] N. Cabot, D. Lamouchi, Y. Yaddaden, and R. Cherif,

- “Drowsiness detection through yawning and eye blinking models using convolutional neural networks and transfer learning,” 2024, csv_key=WF8X73VC.
- [44] F. Makhmudov, D. Turimov, M. Xamidov, F. Nazarov, and Y.-I. Cho, “Real-time fatigue detection algorithms using machine learning for yawning and eye state,” *Sensors*, 2024, csv_key=MMQCZ4P7.
- [45] B. Zhang, D. Yan, X. Wang, L. Wang, and J. Mo, “Fatigue driving detection based on fusion of eeg and blink potential features,” *Journal of Railway Science and Engineering*, vol. 22, no. 8, pp. 3738–3747, 2025.
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